IBU biathlon World Cup points and prize money distribution analysis

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Abstract

This report analyzes the IBU biathlon World Cup points and prize money distribution and compares the FIS cross-country World Cup prize money distribution to the respective IBU version. The analysis tells that points distribution is fairly good and IBU is doing better job on distributing money even with the advantage of having more to share. In both aspects there are still room for improvement.

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1 Introduction

Cross-country skiing is a way of movement in the snow with skis and poles. It's separated from alpine skiing where the skis are used to only come down the hills or mountains. It's origins are in historical times with people who have been living in the areas of snow cover during the periodic cycle. These people have been using wooden skis to move in snowy terrain [11]. In current day cross-country skiing is seen as popular winter activity especially in Nordic countries but it's also a competitive sport. It's been part of the Olympic Winter Games from 1924 in Chamonix [12].

Biathlon is a winter sport which combines cross-country skiing and shooting. It's roots are in Scandinavian survival methods but it has slowly grown as a spectator sport through military competitions. It was part of Olympic Winter Games in 1924 in Chamonix as a Military Patrol and slowly started evolving towards the sport which it's now after 1950s [4].

In the highest international level, cross-country competitions are part of International Ski and Snowboard Federation (short FIS). It includes Winter Olympic Games, FIS Nordic World SKi Championships, FIS Cross-Country World Cup, Tour de Ski, Ski Classics series and more [1]. Basically FIS controls all the major skiing sport events apart from biathlon.

In current day biathlon has it's World Cup tour which is running every year and World Championships which are held on every year other than Winter Olympic years. International Biathlon Union (short IBU) is in charge of the yearly schedule of IBU World Cup and IBU World Championships and they also decide the competition locations and formats. IBU also organizes IBU Cup which are meant is "tier 2" international competition circuit underneath IBU World Cup and IBU Junior Cup which provides international circuit for youth athletes. In the IBUs yearly calendar there are also IBU Open European Championships which is the highlight event of year for the IBU Cup circuit, IBU Youth Junior World Championships and IBU Junior Open European Championships for young athletes and they also provide Para Biathlon events for disabled athletes [6].

In this analysis the main focus will be on events under the umbrella of IBU. It includes the point distribution analysis and comparison between the genders and comparison of prize money distribution between IBU ja FIS World Cups. The goal is to take a deeper dive behind the results and also discuss the prize money distribution and it's healthiness towards the athletes competing in the sport.

2 IBU World Cup competition formats

Over the years the competition formats have changed remarkably but for the last years the format has been fairly constant. The season 2025-2026 consists of the following race formats:

Sprint (7 races)
Pursuit (7 races)
Individual (3 races)
Mass (4 races)

Relay (5 races)
Mixed Relay (3 races)
Single Mixed Relay (3 races)

[9]

The normal format for the season is that there are three (3) competition weekends in a row in three (3) locations and after that there is a longer break. These are called trimesters [10]. Between second and third trimester on each year there is either IBU World Championships or Olympic Winter Games.

For the each competition weekend there are usually three (3) competition days for each gender which are distributed into 3-5 days depending on the weekend. All the competitions use skating style of skiing. Mixed relays are competed always in the same day and usually athletes will be competing in one (mixed or single mixed relay) race but for the smaller countries some athletes might be participating on both. Sprint, pursuit, individual and mass disciplines are counted towards Total Score (explained later) and sprint, individual and relays have separate point system which are counted towards Nations Cup. Nations Cup is not touched in this analysis but it's important for the countries and it determines how many athletes each country is eligible at naming for each competition for the next season IBU World Cup. There are also prize money awarded in each competition [6].

IBU World Cup in the season 2025-2026 contains total of 21 individual competition and 11 relays [10]. The following chapters will explain separate competition formats more closely based on information in [5].

2.1 Sprint

Sprint competition is the shortest of the individual competitions in the circuit. Women will ski total of 7,5 kilometers and men total of 10 kilometers. The race contains two (2) shootings, one (1) prone, one (1) standing, and equal amount of skiing in between of the shootings. The athlete will try to shoot five (5) targets with five (5) shots on each shooting and for every miss they have, they have to ski extra 150 meter loop right after the shooting before continuing for the next lap of skiing. The winner is the fastest to the finish line and the results of the sprint competitions determines start times for the pursuit competition.

Sprint competition can be seen as the most vital competition form of the current circuit. It is competed the most of the individual competitions, all the eligible athletes are open to participate on it and the results will have significant effect on the next competition, the pursuit, which is always competed in the same weekend with the sprint competition. It also counts towards the Nations Cup. Usually sprints had over 100 athletes participating on each time it was competed in the season 2025-2026 for both genders.

2.2 Pursuit

Pursuit competition is usually competed as a pair with sprint as the sprint competition determines the starting intervals for the pursuit competition. Only the top 60 athletes of the previous sprint competition are allowed to participate in the pursuit. In the pursuit competition women ski total of 10 kilometers and men ski 12,5 kilometers. It contains four (4) shootings, two (2) prone, two (2) standing and equal amount of skiing in between the shootings. The each individual shooting work as the same as in sprint. For every miss the athlete must ski extra 150 meter loop before continuing for the next skiing lap. The winner is the first to finish the competition.

In season 2025-2026 the pursuit competition followed the sprint every single time and you could see them as pairs of competitions. In earlier years pursuit has been sometimes competed also after individual competition where the results individual have been the factor of the starting intervals although the time differences have been halved. It can be more tactical format from strategy point of view and also the effect of missing the shots are much more visible during the competition. It's also very dependent on athletes capability of competing in sprint because bad results in sprints will almost quarantee to losing the opportunity to do well in the pursuits. But there are big jumps and big losses of positions seen on every pursuit which means that almost everything is still possible.

Pursuit doesn't count towards Nations Cup.

2.3 Individual

Individual competition is the closest to the original military formats as it's the longest competition from them all. Women ski total of 15 kilometers and men total of 20 kilometers. Once again the race contains four (4) shootings. For the individual race the shooting style alternates in between of the shooting so the order is prone, standing, prone, standings. Each athlete is still shooting five (5) targets with five (5) shots. For every miss athlete will get a penalty minute for their overall time. The winner is the athlete who finished the race in fastest time.

This competition format can be seen as the original format and the name also suggest it being a single individual competition in the history. It's space in the calendar has been diminished for quite some time and some years the IBU World Cup has contained only two (2) individual races for the whole year. For the season 2025-2026 it was raced three (3) times although one of the races was shortened version where skiing distance was lowered and penalty for the miss was only 45 seconds. It's the other individual competition which is counted towards the Nations Cup and countries can place the athletes in the competition which are eligible. For this reason this competition also averaged around 100 athletes per competition in season 2025-2026.

2.4 Mass

Mass competition is final individual competition form. Only 30 athletes can participate into this competition format at one time. In the weekends when it's raced, the top 25 of

Total Score are eligible to participate and 5 of the best athletes outside of top 25 registers their eligibility by their performances during each weekend. There is also a reserve list to fill up the spots which might not be filled to make sure that the competition is always ran with 30 athletes.

Women ski 12,5 kilometers and men 15 kilometers. This format also includes four (4) shootings, two (2) prone, two (2) standing. Individual shootings are also similar and for each miss the athlete must ski the 150 meter extra loop. The winner is the first to the finish line.

This race can be seen as the ultimate goal for the athetes to fill their competition calendar as it contains the smallest amount and the best athletes at each given time. Since it covers the best athletes starting from at the same time, it makes the competitions very exciting and tight until to the end. This competition doesn't count towards the Nations Cup.

2.5 Relays

IBU World Cup has integrated relay competitions tightly into their schedule as there are total of 8 relay days in 9 weekends. It's important for countries to participate to the relays as all the relay competitions counts towards Nations Cup. For this reason in each competition there is an average of 20 countries participating. None of the countries are allowed to set two teams into the relays.

Men's/Women's/Mixed relays work in same way. There are four (4) legs with one (1) athlete competing on each leg for the team. The skiing distance varies but there's always two (2) shootings, one (1) prone and one (1) standing. Specialty in shooting on relays are that each athlete are allowed three (3) extra bullets per shooting in case they miss any of their first five (5) shots. If after eight (8) shots they still haven't hit all the five (5) targets, they have to ski the extra 150 meter loop for each missed target left. In mixed relays the starting order changes for each competition. For every other competition women's legs will be competed first and men will finish the competition and the next one the positions will switch.

Single mixed relay is the newest edition but it's still been competed at least on last six (6) seasons and it has taken it's place in the same day with mixed relay. In single mixed relay one (1) woman and one (1) man athlete will do two legs each. Each leg contains short distance of skiing and two (2) shootings, one (1) prone, one (1) standing. Shooting rules are similar as in the relays. For first three (3) legs the change will happen right after the shooting and possible penalty loops have been finished. In the last leg there is one extra loop of skiing before finish line as in every other competition.

All the relay competitions are competed as mass start competitions and the winner is the first in the finish line. Mixed and single mixed relay are raced during the same day and single mixed relay was added so more athletes could compete in mixed relay day. Mixed relay was added to give smaller countries a change to compete in relays and later adding the single mixed relay gave this chance to countries which won't possible have even two athletes per gender competing in IBU World Cup circuit in regular basis. The investments

of IBU for helping smaller countries in more ways than adding these competitions is seen especially in single mixed relays as in some weekends the list of starting countries are getting close to the max of 30 countries.

3 FIS Cross-Country competition formats

Over the years in FIS Cross-Country World Cup the schedule have been containing wider variety of competition formats than IBU World Cup. For couple years and also in season 2025-2026 FIS Cross-Country World Cup schedule has been more streamlined and cohesive. In general there are three (3) different distances for skiing races with mix of two (2) different skiing styles or sometimes having both of them in one competition. There also also team sprints, which are two athlete team competitions but there are no other relays on the season schedule. Due to the two (2) skiing styles there are fairly low amount of identical races for the year as all the three (3) distances are split into half [2].

In general there are usually slightly less athletes competing on each event compared to the biathlon. In women's competitions there are usually between 50-70 athletes with some cases having slightly more or slightly less. In men's races there are usually between 70-90 athletes. In total FIS cross-country World Cup contained 27 events, 16 distance events and 11 sprints. In distance events there was also bonus points awarded for the leaders of the race in certain parts of the track which counts towards overall standings [2].

The following chapters explain different competition formats citing the official FIS website.

3.1 Sprint

Sprint is maybe more of what the name suggests than a biathlon sprint. The sprint skiing is done usually in a track which is around 1,5 kilometers long. Depending on the track and conditions the goal time frame is between 2 minutes and 30 seconds and 3 minutes and 30 seconds. Sometimes, for example often in the Olympic Winter Games and World Championships, the track might be slightly longer and more challenging. Both women and men are racing the same track. This competition format is also ran with two different skiing styles: classic and skate.

In this competition format there are two phases: qualification and group stages. In qualification all the participating athletes will ski around the sprint track as fast as possible with 15 second starting intervals. Based on these times top 30 athletes will be qualified into the group stage. Based on the standings and the picking rules the athletes can pick their groups for the group stage.

The group stage is done in groups of six (6) athletes. From each heat two best will qualify into the next stage and 2 non-qualified athletes with best times will qualify as lucky losers. In group stage there are total of three (3) stages: first round (5 heats), semi-finals (2 heats) and final. Winner is the fastest to the finish line in final and positions outside the final are decided based on 1) the finishing position in the heat and 2) time.

3.2 Interval Start

In interval start competition format each athlete will start in 30 seconds intervals. In earlier season it varied how long the competition distance was but for now interval start competitions are always 10 kilometers long. The winner is who finishes the distance in fastest time. The competition format is ran with two different skiing styles: classic and skate.

3.3 Mass Start

A mass start format is very much similar to biathlon mass start format except there is not similar limitation for participating which means that everyone who are allowed to start in World Cup competitions can participate. The competition distance on this format is currently 20 kilometers. The winner is the fastest to the finish line. It's ran with three different styles: classic, skate and skiathlon. Skiathlon is a combination of the both styles. The competition starts with 10 kilometers with classic style, then all the athletes change their skis and the competition is finished with 10 kilometers with skating style.

3.4 Special events

During the year there's also Tour de Ski, a special combination of competitions which counts as a separate tour but also counts towards the World Cup but with modified points and prize money distribution. It's schedule varied year-by-year and it contains the regular distance competitions but also sometimes there are more special competitions in terms of distance and format. Tour de Ski is held every year around new year.

Other specialty is a Holmenkollen 50 kilometers competition which stands out from the rest of the schedule by it's distance. It has legendary status and when the World Cup schedule was revamped to it's current form, this competition maintained it's position in a calendar in it's original form. The only change for this is that also women are now racing the 50 kilometers as they used to ski 30 kilometers before.

Cross-country is also part of the Olympic Winter Games, which means that in season 2025-2026 it was the major event in the calendar. There is a differentiating thing compared to biathlon calendar however as the cross-country World Championships are held only once in two years. This means that every fourth year there isn't a "major" event in the calendar. Usually in these years there is more emphasis on competing in Tour de Ski.

4 IBU World Cup ranking systems and prize money distribution

In the IBU World Cup circuit there are running several different ranking systems parallel to each other. IBU also awards the best athletes/teams of the each competition with prize money. The summary of the explanation can be found here [7].

Total Score is the individual competition for the whole year which determines the best athlete of the IBU World Cup each year. It contains all the individual competitions, 21 in season 2025-2026. It's the most visible part of the competition circuit. Total Score point distribution saw a major reshape 4 season ago and has been tweaked last time for the current season 2025-2026. For each race top 40 athletes will collect points.

Nations Cup is the other ranking system which is meant for the countries. It's main purpose is to determine the starting spots for the next year for each country but the best countries are also awarded with prize money based on their performance. Nations Cup is not looked into more in-depth in this analysis.

On top of these each individual discipline is also followed separately with the same points distribution system as Total Score. They also have rankings for relays and mixed relays. Winner of all of these disciplines are awarded at the end of the season.

IBU also awards best teams with prize money for each race and there are some end of the season bonuses and other bonuses as well. In each individual race top 30 athletes will get awarded with prize money and in relays top 8 countries do as well. At the end of the season top 10 best athletes will get a bonus and winners of each discipline will also get a prize money bonus.

4.1 Ranking points distribution

Total Score and individual disciplines follow the same points ranking distribution. Total of top 40 athletes will get points from each race. For the mass format the point distribution is modified from the bottom to fit better with 30 athlete race. The distribution has changed 4 season ago when it was decided that top position will deserve some more points on their success compared. Before this change the winner did get 60 points, second 54 points, third 48 and so on. In the new system winner will get 90 points and second 75 points but the rest of it has been tweated on last couple years. Currently the top 10 will get following points:

Position	Points
1	90
2	75
3	65
4	55
5	50
6	45
7	41
8	37
9	34
10	31

From this forward each position is worth of one points less until 40th will get one (1) point (expect in mass). Positions from 10 forwards has remained the same for the whole

change period so technically it's possible to compared seasons from athletes without any modifications for decades back if they have not finished any times in top 10.

4.2 Prize money distribution

IBU has also contributing for the athletes success by awarding them prize money in various ways. Total amount of prize money awarded out for the athletes in various cups is roughly 8,5 million euros. In IBU World Cup there are multiple performance based awards for each race and also so called bib-bonuses for starting as a leader of different competitions.

The main way to gather prize money with successful performances is to finish in top 30 in individual races or top 8 in any relays. In the season 2025-2025 winner of individual race is awarded with 20 000 euro prize and award declines down to 250 euros which is given to 30th finishing athlete. In relays winner team gets 30 000 euros and the last awarded place of 8th receives 2 400 euros. For single mixed relay this the relay prize money is halved due to team consisting only half of the members of a normal relay. In this way the individual of the team will receive always same prize money from same finish in any relay competition.

In the years when Olympic Winter Games aren't organized, IBU organizes IBU World Championships where the prize money from individual races are slightly higher than from normal IBU World Cup race. For the season 2025-2026 each IBU World Cup race was worth closer to IBU World Championships as the prize money reserved for that was shared in the IBU World cup. For example in previous years the individual competition winner has granted a 15 000 euros for their success and the rest of the distribution was adjusted accordingly.

On top of these there are some other bonuses. The main ones are the end of season Total Score result bonus which is awarded to top 10 athletes in the points. Winner of each smaller cup (sprint, pursuit, individual and mass) will also receive a bonus award. There is also bonus award for the best U23 athlete and nations will get a bonus from winning relay and mixed relay cups and performing well in Nations Cup.

But there's more. During the season there is yellow bib for the Total Score leader, red bib for the each discipline leader and blue bib for the best U23 athlete in Total Score. The athlete who happens to lead these competitions and is starting to the next competition is awarded a "bib-bonus" for each start they wear these bibs. Theoretically it's possible to wear all these bibs and then you get all the bonus start awards from that specific competition. Newer additions are also bonus awards for fastest pursuit time and fastest leg on relays and best woman/man in single mixed relay which combines the two legs each are competing in each race.

The whole 8,5 million euros doesn't go the top level. Secondary level IBU Cup and it's main event IBU Open European Championships are also getting their fair share of the prize moneys. This and the whole prize money distribution and amounts are covered more in-depth in IBU website [8].

5 FIS Cross-Country ranking system and prize money distribution

As for IBU World Cup, in FIS cross-country World Cup has also several different ranking systems running parallel to each other and it too has prize money for the best athletes. The goals of these systems are more or less the same but they some some slight differences. As for this analysis the ranking systems are not used for comparison, they will be introduced only briefly. Prize money distribution is covered slightly more in detail. The more detailed information for this can be found from here [3].

5.1 Ranking points systems

The main competition is the overall cup which includes all the World Cup competitions and theirs results. The winner is the athlete who collects the most points during the year. From each individual competition total of 50 athletes will get points. FIS has made a change to the points distribution at the same time with IBU but they made the decision to go opposite direction. Win has always been worth of 100 points, but currently the places are worth 95, 90 points etc. as before the difference was much larger with 80, 60 points etc. There is also Nations Cup but as there are not as much competing countries and it's implementation to the tour isn't quite as the same at in IBU Nations Cup, it mostly flies under the radar.

5.2 Prize money distribution

As in IBU in biathlon, FIS in cross-country contributes to the best athletes with money prizes. In the season 2025-2026 around 4 million euros was awarded to the athletes, including the FIS cross-country World Cup and Tour de Ski. The there are multiple differences but one of the major ones, which requires mention is that there isn't a full tier-2 level international competition circuit. The next step underneath are few regional and national cups with less coverage over the year.

As for the biathlon, the each World Cup race has it's own prize money pool. Top 20 athletes in individual races and top 10 teams in team sprints will be awarded with prize money. In the season 2025-2026 the winner of the individual race will get 15 000 euro prize and the last prize awarded for the 20th place is 500 euros. In team sprints winner team will get 16 000 euros and 10th 1000 euros. All the other prizes are in between.

On top of this at the end of the season there is prize money awarded based on the overall standings. Also here top 20 athletes will be granted some prize money ranging from the winner, who receives 55 000 euros to 20th, who receives 1000 euros. In Tour de Skis independent events only top 3 will get a smaller award but for top 20 who finish the whole tour will get awarded starting from winner, who gets 80 000 euros. There's also couple minor bonuses for top athletes in best sprinters and best climbers.

6 Pre-analysis prize money distribution comparison

Before analysis part of the report it's good to highlight some key factors and compare these two distributions with each other.

First, it's important to mention that in IBU World Cup the Olympic year was a special year. In usual year for the recent years, the winner has been awarded with exactly the similar amount than what FIS cross-country World Cup winner receives in this year. As in a usual year IBU will also award prize money in the World Championships the individual competitions receiving the prize money is roughly the same, unlike this year. FIS doesn't award any prize money in their World Championships. It's still good to point out that in a usual year in IBU events on top of the individual competitions there are roughly 10 relays which athlete can participate to gain prize money when in FIS events there are only couple.

The main difference between the sports is that in biathlon there is over double the prize money which is awarded over the year. Partly due to this reason it wouldn't be useful to compare these with each other cause in most cases throughout the whole athlete pool the biathlete will receive more prize money compared to the same ranked skiing athlete. This is also why it's better to compare the prize money distribution in percentages of the whole prize money awarded during the year and look if there are differences in the spread out between these sports. To achieve same amount of money, the athlete in the similar position in the ranking should receive almost twice the money from whole pool in FIS World Cup than in IBU World Cup.

From technical point of view there are also major differences. In biathlon in every individual race 10 more athletes will be awarded which in theory should distribute the money for more wider group of athletes. In cross-country in team events more teams are getting prize money, although in the end it doesn't cover as many athletes than relay days in biathlon (20 athletes in team sprint per gender in cross-country vs 24 athletes in mixed relay days or 32 athletes in relay days per gender in biathlon). The bigger prize money pool gives also opportunity for biathlon to share bigger prize moneys for lower finishing athletes in the race. In season 2025-2026 if you placed on 6th place in the individual competition (in 2024-2025 5th place), you got awarded more prize money than if you finished 3rd in cross-country. This continues mostly throughout the whole distribution but the biggest dropout in the FIS World Cup is definitely after the winner and 2nd place. The fact that less athletes are awarded prize money per race and the drop is way more drastic it should mean that less athletes should get prize money.

The other differences comes at the end of the season bonuses. In FIS cross-country World Cup winner is awarded more prize money but it also awards much wider range of athletes, although with much smaller prizes. The single most valuable event to win is still Tour de Ski, which winner receives 80 000 euros worth of prize money. This distribution is also fairly top heavy but it's still significant as 10th is still receiving 9 000 euros. It's good to mention a certain issue of the distribution which is that individual Tour de Ski events will only grant prize money to top 3 athletes. As there was six (6) competitions part of the tour this year, finishing 10th granted less money than finishing six (6) times as 10th in normal

FIS World Cup competition. Also IBU awards the discipline winners with prize money at the end of the season, which FIS doesn't do.

7 Analysis

The goal of the analysis part is use visualization as a tool to understand and make observations from the results. Analysis is will cover IBU World Cup point distribution analysis and comparison and prize money distribution comparison.

The information is gathered from IBU Datacenter and official FIS website. Some of the features are available straight from the source and in the case of FIS related data it has been used without further modifying. with IBU related data the straight collection of data didn't give the opportunity to do more much so all the data from biathlon races are gathered from different parts of the same source, modified and combined for the purpose in hand. For example when official FIS website covers the prize money distribution of all the athletes in the season 2025-2026, official IBU website didn't do this and this had to be done by modifying the competition data. Doing more data preparation also gives opportunity to go further with the analysis covered in this report such as analyzing the changes in points and prize money distribution or seasonal comparisons.

7.1 IBU Women's Total Score race

The main discipline in the IBU World Cup is the individual overall competition called Total Score. As mentioned before, it contains 21 individual races in season 2025-2026. From the official IBU website we can gather that the winner of this years women's Total Score race was Lou Jeanmonnot from France [9]. She gathered total of 1135 points and had fairly big margin to the others. During the season total of 172 athletes were competing at least in one race and 102 out of them managed to finish at least once in top 40 to gather some points.

First look will be the distribution of the points from top 30 athletes from the season 2025-2026 divided into two graphs and at the end we take a look into development of top 10 athletes and how they ended up into their final results. All the specific information about competition results can be found through [9]. If there is a mention about a specific athletes competititon results, you can find them by clicking their name to open their profile and look for latest results. Some statistics mentioned are only available through the data processing.

Graph (1) shows top 15 athletes of the season. The graph shows fairly big gap between the first and the rest but there is another gap after top 7 athletes suggesting that these athletes were the ones competing from the highest position with the highest consistency. Based on points distribution on single events and compared that to the total points, the rest of the athletes had fairly good chance on fighting from top 3 positions but they couldn't do it consistently. Either their consistency or overall level was one step behind top the top 7 athletes.

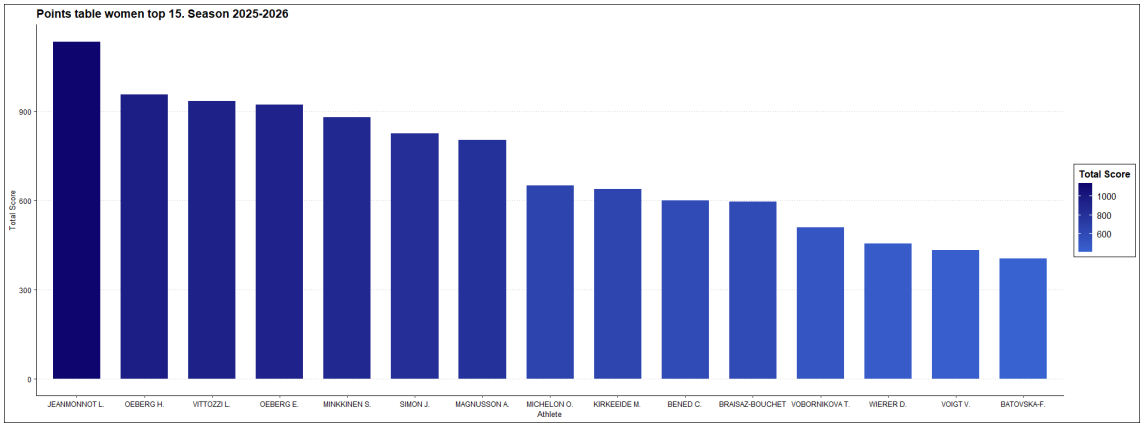

Figure 1: Points from women's top 15

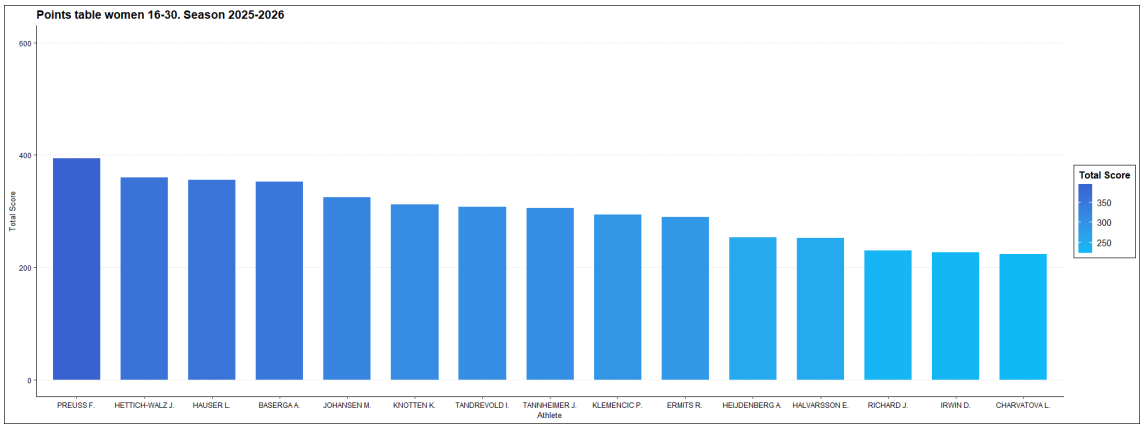

Figure 2: Points from women's top 16-30

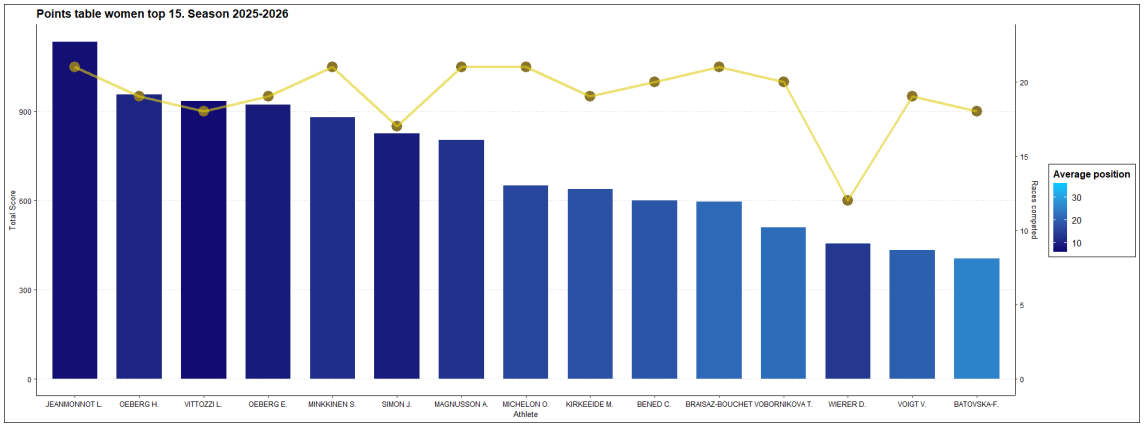

Figure 3: Points from women's top 15 with added gradient based on average position and dots and line showing the amount of races competed.

Graph (2) shows the next best 15 athletes. Just as expected the gaps between the athletes starts to became smaller. Most of these athletes did have chance to surprise in single race, which could mean top 6 finishes in their case, to gather bigger points but they didn't have the same consistency for fighting from the absolute top positions throughout the season than the group above on positions 8-15. Greatest example of a big suprise from this group would be Lisa Hauser winning a Östersund Pursuit after finishing 4th in the preceding Sprint competition. Including both races, she didn't miss a single target, which she wasn't able to reproduce in the rest of the season.

To make analysis more in-depth some extra features should be added to understand the results better. To do this for the next two graphs the gradient of the bars have been switched from points to average position of the athlete throughout the season. For more information a line graph with dots was added to showcase how many races each athlete participated in the season 2025-2026. Y-axis of the races competed can be seen on the right side of the plot. To make analysis possible in between of the graphs the range of the gradient scale has been adjusted to be the same. The darker color signifies the higher average position.

The changes in the gradient in the graph (3) are fairly subtle but there are some things to gather. When combining all the three information, total points, average position and races competed, it's possible to make some observations. Clear outlier in the top 15 is Dorothea Wierer who finished her career after Winter Olympics and didn't compete in the last trimester. Due to this fact she didn't participate as many races than others in average but she had a very good season with clearly higher average position than athletes around her. Other than this outlier the top 7 separates from the rest of the group while having subtle differences. In general it's possible to say that if athlete competed more often, she had slightly weaker average position compared to the athletes finishing close to similar points. Also it's good to mention the winner Lou Jeanmonnot reaching a average position of 7.2 which also wasn't quite the highest which came from Liza Vittozzi with 6.8.

From the second graph (4) it's possible to find another clear outlier but possibly also another smaller outlier. Franziska Preuss had similar decision to Dorothea Wierer and she also finished her career in the Winter Olympic Games. Preuss was performing even

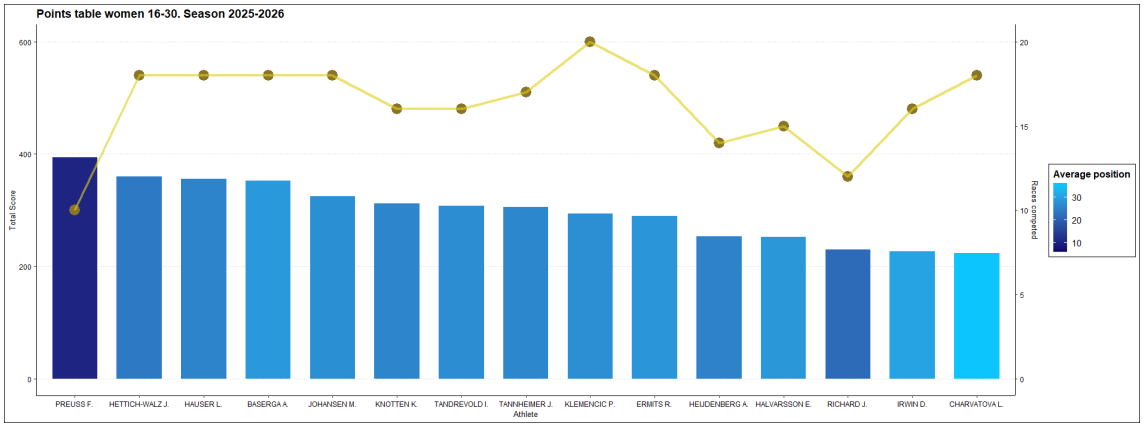

Figure 4: Points from women's top 16-30 with added gradient based on average position and dots and line showing the amount of races competed.

higher standard as her companion, although Wierer had won a competition which Preuss didn't achieve, but on top of the missing third trimester she also missed some competitions from the start of the season. In total she did compete in less than half of the all individual competitions during the season. If competing for the whole season, she was clearly capable of competing in the level of top 7 athletes, which Wierer was able to do as well but with higher inconsistency. Based on this analysis they can be added into the same group with top 7 athletes to create the top 9 athletes who separated from others.

Before other outlier the colors are signifying what were expected already based on points. Most of these athletes 4 weren't able to compete consistently from the top positions, which shows as higher average position per race. These athletes also had few less competitions on their belt during the season compared to top 15. The reasons may vary from health issues, preparation to Olympics, missing mass starts or in some cases there could be another issue which French athlete Jeanne Richard was facing. She is the another outlier in top 16-30 and she had problems to fit into French team due the restrictions of 6 athletes for their country. The competition from these spots was very high as France is currently having the most amount of high level athletes and they all can't fit into the World Cup competitions at the same time. She was performing better than athletes around her but she was forced to give her spot in Annecy and the whole last trimester for other athletes. If competing the whole year she could've been part of the top 20 athletes of even higher.

Looking into the development of the season from top 10 athletes 5 it's possible to see couple different progression styles. There were couple early season successors, including Anna Magnusson and Suvi Minkkinen, who claimed the top position in Total Score mostly during the first steps into the season. Lou Jeanmonnot managed to overtake them already still during the first trimester and claimed majority of her lead in the second trimester compared to the others. Hanna Öberg did have a good result on very first competition but had slightly slower mid season until the end of third trimester where she managed to overtake 4 athletes in last 3 competitions to claim second place in the standings. Julia Simon was the late entry for the season starting only the second weekend but third trimester was the best for her. Missing the first weeks action did drop her slightly behind from second to fifth place. Oceane Michelon had a stable year but her results were slightly weaker than

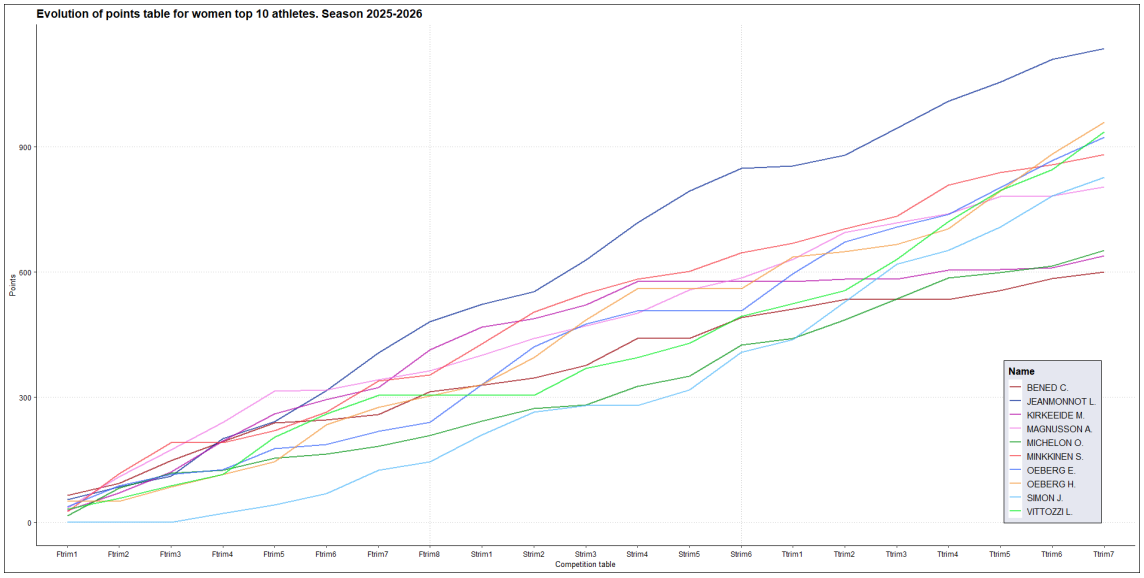

Figure 5: Development of the points table from the top 10 women athletes at the end of the season 2025-2026.

the most until the end where she managed to overtake two athletes.

Visualizing the season like in graph 5 can give even person not following a peak to how season progressed. After first trimester it looked like athletes like Maren Kirkeeide, Magnusson and Minkkinen, who were new names for the fight from overall win would be competing the hardest from it against the possibly biggest favourite Jeanmonnot. Meanwhile many other star names, who ended up pushing to top 6 at the end of the season, would be missing out and they probably did. The third trimester did show that with having a great success during one of the three trimester can still push you near to the top but if your season is quite not consistent on high level, winning the overall is nearly impossible. It also showed that a lot of the biggest stars did prioritize the Winter Olympic Games over balanced performance in the IBU World Cup and this is also maybe a reason why Jeanmonnot was able to win the overall with comfort while maybe not performing not as high level than year before.

7.2 IBU Men's Total Score race

As for women, also men had 21 individual races for the season 2025-2026. Men's winner is Eric Perrot and comes also from France [9]. His gap for the rest of the athletes was even bigger than in the women's side. In men's race there was some more unfortunate events during the season other than skipping some IBU World Cup weekends to prepare for the Olympic Winter Games or athletes retiring before the season ending. A surprise death of an athlete during Christmas break and serious heart problems effected on handful of top athletes to skip some of the competition and most likely the effected the Total Score race in major ways. Credit has to be given to the winner Perrot even so, as he was super consistent in the top level over the whole season, which will be covered in a moment. Again, the personal competition results can be found from [9] but some statistics are only available through the analysis.

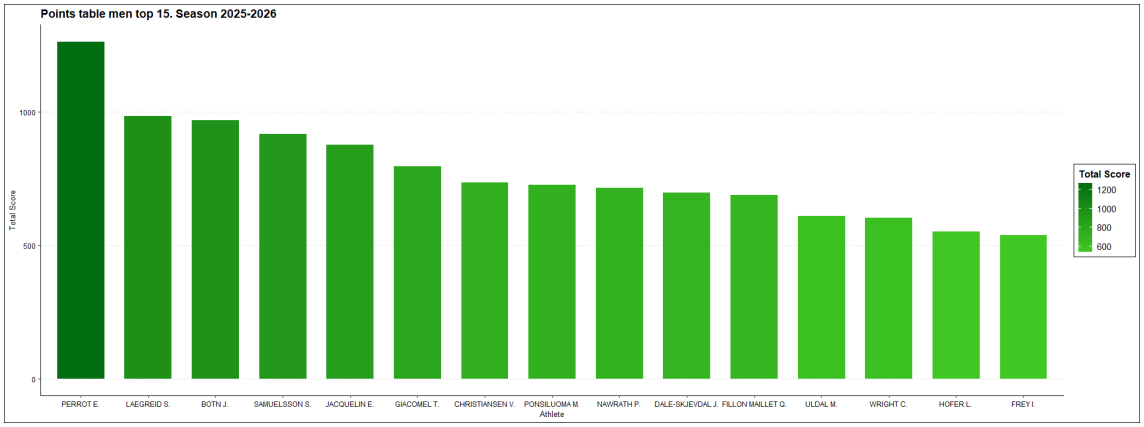

Figure 6: Points from men's top 15

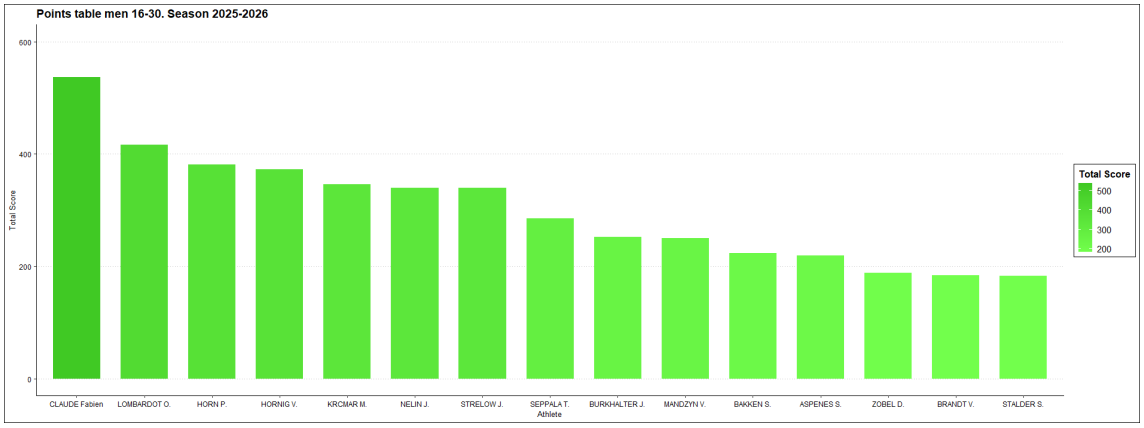

Figure 7: Points from men's top 16-30

In the season 2025-2026 total of 187 men competed at least once and total of 105 of them manage to finish one or more times in top 40 to gather points for their Total Score results.

In the first graph (6) the top 15 athletes on men are introduced. In men's competition it's not quite as obvious to notice a clear gap in points to create groups from the athletes inside the top 15. Interpretation could be to separate top 5 athletes from each other but Italian Tommasso Giacomel is sort of in between of the two groups. It's clear that more information is needed for better analysis.

Looking into second graph (7) and this information to what was in the first graph (6) the clear gap between athletes is now visible. After Fabien Claude in 16th place there is clear gap between points. The rest of the athletes are more of less in the pack although it could be argued that since the point distribution is smaller after top 10 on each race that after Claude there's another group of 6 athletes and a gap for the rest of the athletes as well. Based on knowledge outside of pure data crunch, this gap basically is created as the top 22 athletes were able to gather at least few top 10 positions which were more valuable with point distribution. There are of course few exceptions to this rule which will be mentioned after improving the visualization.

To improve analysis the gradient is changed to the average position and the race total is

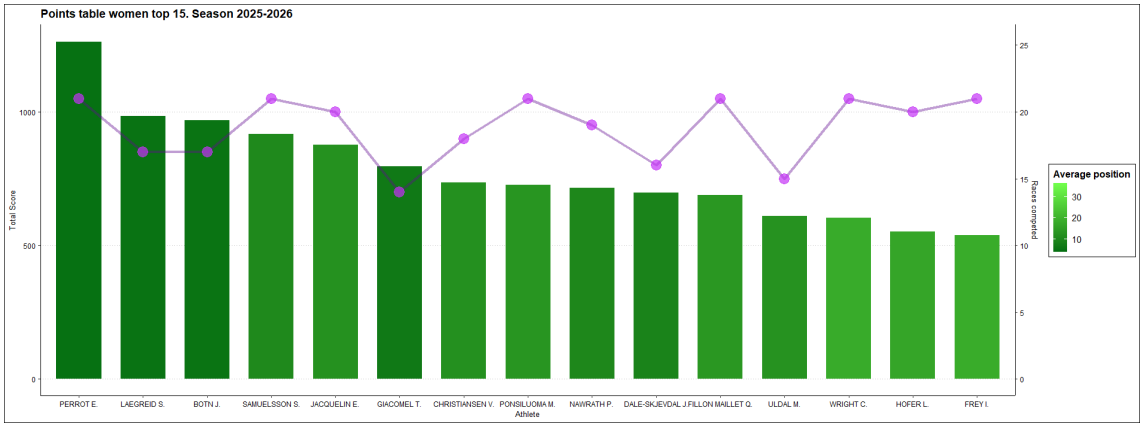

Figure 8: Points from men's top 15 with added gradient based on average position and dots and line showing the amount of races competed.

added to the graph. The gradient is again the same over the two following graphs.

(8) With updated gradient the graph is not the easiest to analyze as the top athletes perform clearly better than the athletes in the graph (9). With closer look it's possible to separate four athletes who were performing better than the rest in terms of average position. These athletes are Perrot, Sturla Holm Lägreid, Johan-Olav Botn and Giacomel. Giacomel out of these four had to skip the last trimester due to the health issues. The main reason why the gaps between these athletes are so big is the amount of competitions they were participating throughout the year.

From the rest of the top 15 it's possible to recognize two more athletes who performed slightly better than their closest rivals and they are Johannes Dale-Skjevdal and Martin Uldal. They also had less competitions on their belts over the season and the reason for this based on further analysis is that Dale-Skjevdal skipped a weekend to prepare for the Olympic Winter Games and Uldal most likely had some issues with her health, leaving him out from one of the weekends at the start of the season. If competing the whole season they probably would've been slightly higher in the final results but still behind the top 4 athletes.

When looking for the rest of the top 30 in (9) the gradient predictably becomes lot brighter. The most notable athlete is Sivert Guttorm Bakken who competed only two weekends and had clearly better success compared to the athletes closest to his points. By side of him there's also another Norwegian Sverre Dahlen Aspenes who also competed only few times during the season and he was facing similar issues than Richard in women's side, national competition. He did also have fairly poor performances and almost half of his points came from two competitions in Nove Mesto.

Looking at the development of the competition of overall in men's top 10 10 it's possible to see how Botn was basically dominating the first trimester, Giacomel took the lead due Botn missing some competitions and performing slightly better in second trimester than Perrot but eventually it was Perrot who took the lead for the end of the season as Giacomel wasn't able to compete. Also Sebastian Samuelsson was close on the race to overall win

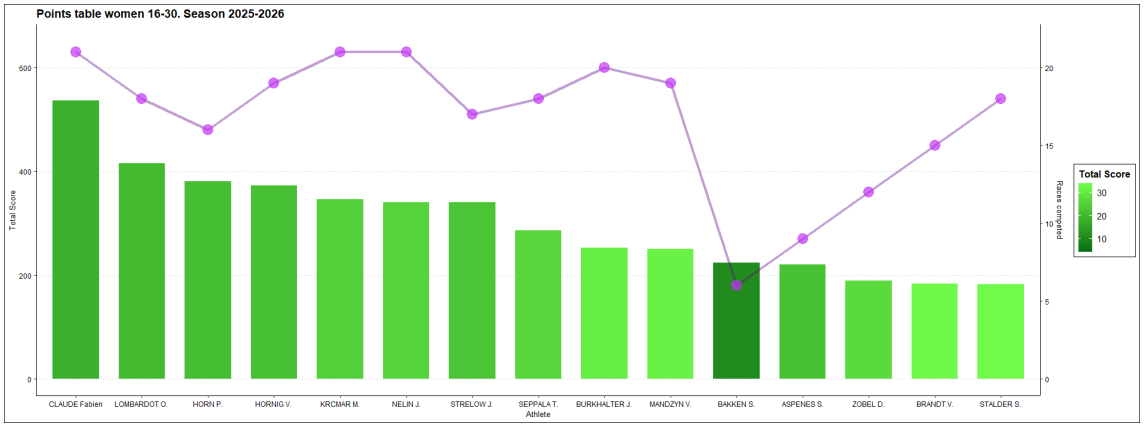

Figure 9: Points from men's top 16-30 with added gradient based on average position and dots and line showing the amount of races competed.

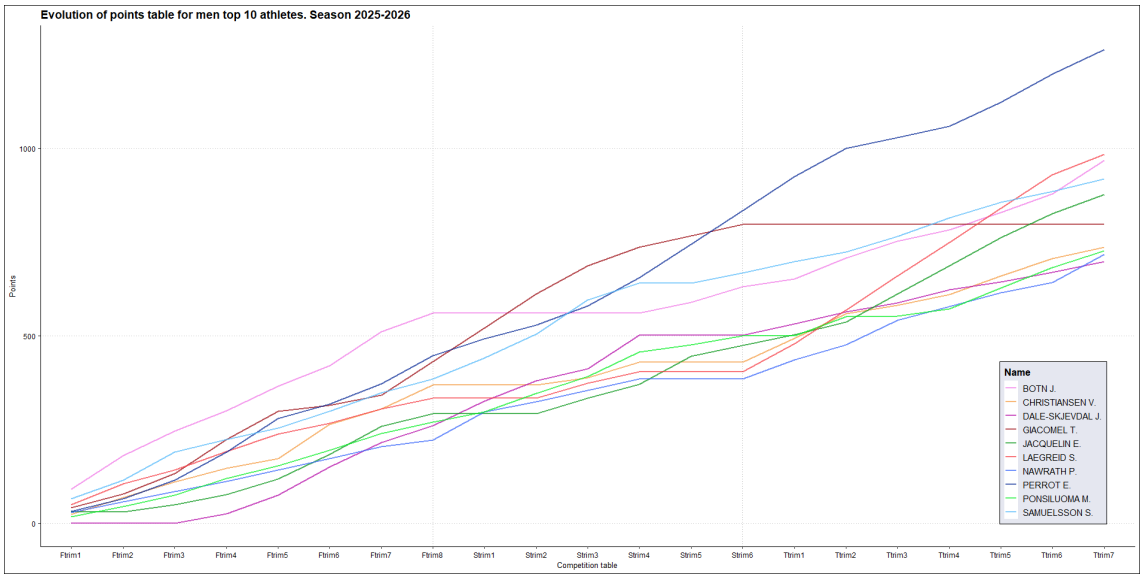

Figure 10: Development of the points table from the top 10 men athletes at the end of the season 2025-2026.

mid-way to the season but wasn't able to keep the level of Perrot at the second half. The rest of the top 10 did perform fairly evenly until the third semester where Lägreid was able to peak from the rest and finished to the second place in the overall by winning all but two competitions after the Winter Olympic Games and finishing worst at fourth. The main reason why he wasn't able to catch the winner Perrot more was because of Perrot's great consistency.

In general both women's and men's graph showcase that usually top 15 athletes had on average more competitions but the margin isn't that high that it would be significant in most cases. Apart from the preparation for the Olympic Winter Games, health issues or not making the team being the reasons to not start, this can also be due to the fact that only top 25 athletes are allowed to start in mass competition and some of the athletes at the bottom of the analysis were not eligible to automatically participate in all the races. It's also more likely that athletes around top 30 in the Total Score had slightly higher chance of missing the pursuit competition after poor performance in sprint compared to the top athletes. In men's side the group of athletes able to compete consistently from top 6 finishes where slightly wider which also created a major gap after this group to the athletes who barely were able to reach into top 10 during the season to gain bigger chuck of points from one finish.

7.3 IBU Points distribution comparison

As the season structure and point distributions are equal between genders, it's possible also to compare the results in between of each other. This analysis can reveal differences in between two competitions and it's also possible doable over seasons. For now the comparison is made from season 2025-2026 Total Score results.

As mentioned earlier, 102 women and 105 men were able to collect points during the season. Total amount of competitors was also relatively close, 172 for women versus 187 for men. For the next graph all the athletes who managed to collect points were included. For additional help two vertical dotted lines were added on 15th and 30th place in comparison to make visibly more clear the comparison earlier made gender specifically.

First observation looking at graph (11) is the obvious one where winner of the men's Total Score race scored more points than women's winner. This was mainly possible because of higher competitive consistency. Other than this fairly clear difference the top athletes were very comparable between each other in terms of points gathered. First major difference in the distribution comes when women's distribution has the first major gap discussed about earlier after 7 athletes. From this point men's athletes systematically were able to collect more points on average up to around 20th athlete. This can be seen as a sign from wider competition in men's side right after the top athletes. This was also possible to see in graphs (8) and (9) where the first clear gap was after 16 athletes. Even this drop wasn't yet enough to make women's athletes to collect more points in comparison but instead the next gap discussed before finally made it.

The second major cut off point is after 20 or so athletes when men's distribution has bigger gap. After this change women's athlete consistently collected more points until the very

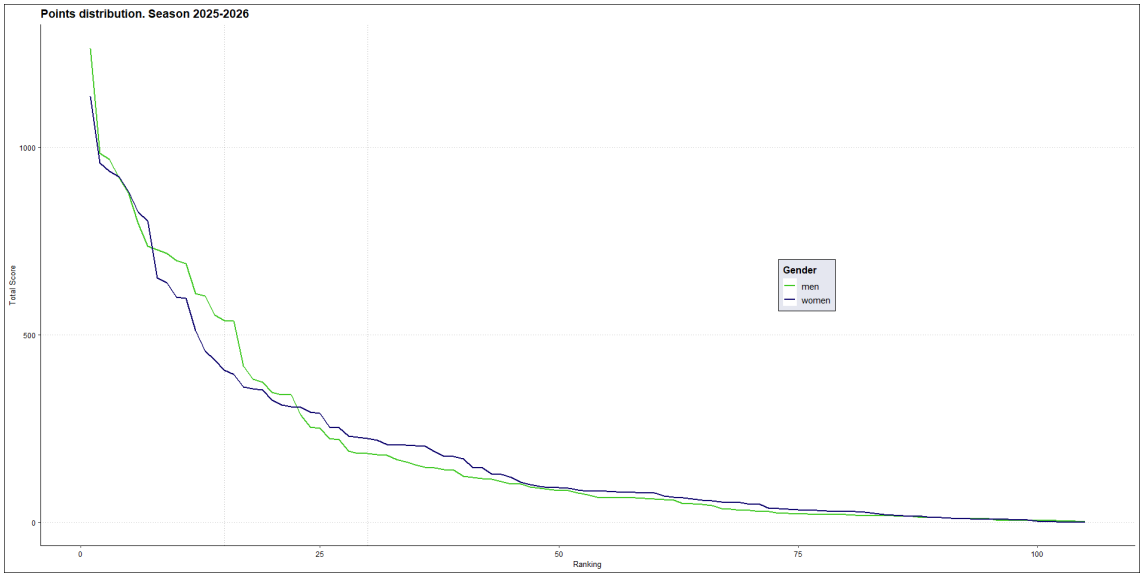

Figure 11: Comparison of point distribution on season 2025-2026.

end of the graph where the last few athletes are represented who manage to collect just few points, although this gap closes in already earlier. This would still suggest that in women's side the mid level competition is more equal than in men's side.

One reason could be also that in men's competitions there are more athletes who are capable for top results. Finishing one time in the podium per season while performing in solid level for the rest of the season, usually results over certain threshold of points, which is difficult to achieve if athlete is consistent in top 20, but can't break into top 6 or podium. This leads into situation where there is less to gain for the rest.

In women's side the competition is maybe not quite as wide in the absolute top which then gives more opportunity for wider group of athletes in the mid range of the pack. In technical terms it still applies that even one or two top performances with solid consistency will pull an athlete over certain threshold but in women's competition there can be more standard deviation for certain athletes. One example of this is Finnish athlete Sonja Leinamo who started the season with excellent second place finish in the first individual competition but after this she could finish to the points more than once in another individual competition. She for example couldn't finish into top 60 in any of the sprints, a hard line she had not cut more than once on season before either. Because of this she is an outlier of one podium performance but not even close of able to be part of out earlier analysis of top 30 athletes of the year.

In general it can be said that there are maybe more of a mix in the women's competitions right after to absolute top and there are also more women who are capable at reaching even top 15. When in men's competition there are more athletes who are able to peak at higher level, it does leave less room for others even when they would do near to perfect competition while others with peak capability would not. One metric that was measured but not very visible in the analysis, was standard deviations of the positions over the year and average of this variable was slightly higher with women athletes than with men

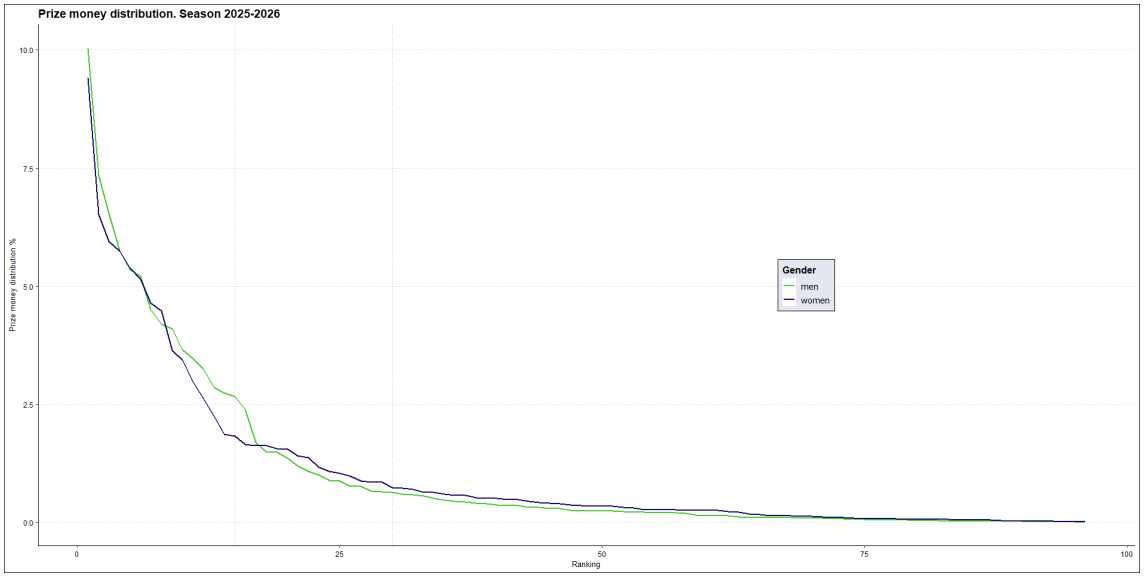

Figure 12: Comparison of prize money distribution on season 2025-2026. Comparison is made in percentages of total prize money distributed for the season 2025-2026 per gender.

athletes. This could in theory suggest the claim but the difference wasn't very high (women 15.4 vs men 14.4).

7.4 IBU Prize money distribution comparison

Next step is to compare prize money distribution. As the prize money is awarded to smaller amount of people it's expected that less athletes will be awarded by money prizes during the year. Total of 96 women and 95 men managed to finish in top 30 or be in a top 8 relay team at least once during the season 2025-2026. It's also not a big surprise that the Total Score winners, Jeanmonnot and Perrot, were able to snatch the biggest portion of the prizes. This obviously expected but in theory in a tight title race year it could be opposite. The main question is how much of the prize money will go towards the top and is it possible to determine a some kind of a limit, how many athletes can theoretically fund their professionalism via prize moneys earned during the year?

As IBU ain't publicly sharing prize moneys won by athletes, it won't be published in this report either for the respect of the athletes and IBU. Calculating the sum is still totally possible to anyone as all the information needed is publicly available. This comparison is made by using the percentages of how much of the total sum was each athlete was able to gather. This sum is gender based so the sum of percentages for women and men athletes respectively are hundred percent. This also gives better way to compare distribution between different sports with different amounts of prize money awarded throughout their seasons. When asking to the second question asked previously, it has to be taken into account that total prize money sums will differ. Looking into percentages can also reveal better the healthiness of the prize money distribution system than comparing straight sums of money.

As seen in graph (12) it follows fairly similar pattern than when comparing points in (11).

It doesn't have as big of gaps which can be partially explained by calculation method but it also shows that even points distribution had bigger gaps, the prize money did even out between genders in fairly similar form.

One main interest of this comparison was to possibly see if there was much of mix in rankings of athletes with their earned points and prize money won throughout the season with different qualities. Interesting information gained while building this visualization during the last weekends of the season was that before the last weekend there were bigger mixes in between of athletes when compared points and prize moneys. After adding the final week and end of the season bonuses it all streamlined and only bigger outliers were the U23 cup winners who basically collected all the blue bib bonuses and won the U23 cup and therefore gained few positions in the prize money ranking over points ranking.

Even the end of the season bonuses did added some more top heaviness to the distribution, it was fairly subtle. The distribution of the prize money throughout the different bonuses is fairly interesting and also raises the question of the what is the reason for each of them. For example is the smaller discipline winner prize money's point to award the person who ends up being the best in the discipline and this way share the prize money to more than one person or is it just a bonus for the Total Score winners if and when they are dominating the whole competition throughout the season? Is it also that with old points distribution method there was slightly more variance with the small discipline winners? Regardless of the reason, on this season both overall winners did also won three out of four smaller disciplines.

The answer to the second question of is it possible to determine a some kind of a limit, how many athletes can theoretically fund their professionalism via prize moneys, the first goal would be to set a limit for what is enough for an athlete to maintain her status as a professional athlete. Setting this kind of a limit is really difficult due the multiple different reasons. There are also multiple sources of funding which are possible for the athletes and usually the sponsor money is also more accessible for the top athletes than mid-level athletes. To give some kind of value to compare with cross-country in next chapter, around 60 women and 50 men were able to break a five figure prize money during season 2025-2026. This counts over one third of women athletes and over 25% of men athletes who participated at least one competition in IBU World Cup this season.

One major point to add is that IBU does have the second tier international competition circuit, IBU Cup, as well which part of the total prize money is distributed. With a quick glance over top athletes, it is confirmed that there are at least handful of athletes who are able to collect a five figure amount of prize money while only competing in IBU Cup. This is good information to take into account when comparing biathlon and cross-country prize money distribution in following chapter.

7.5 FIS vs IBU prize money comparison

Next the FIS cross-country World Cup prize money distribution is added to the graph to compare how the prize money is distributed within each sport. This comparison should give us more information about differences on distributions and can be used to compare

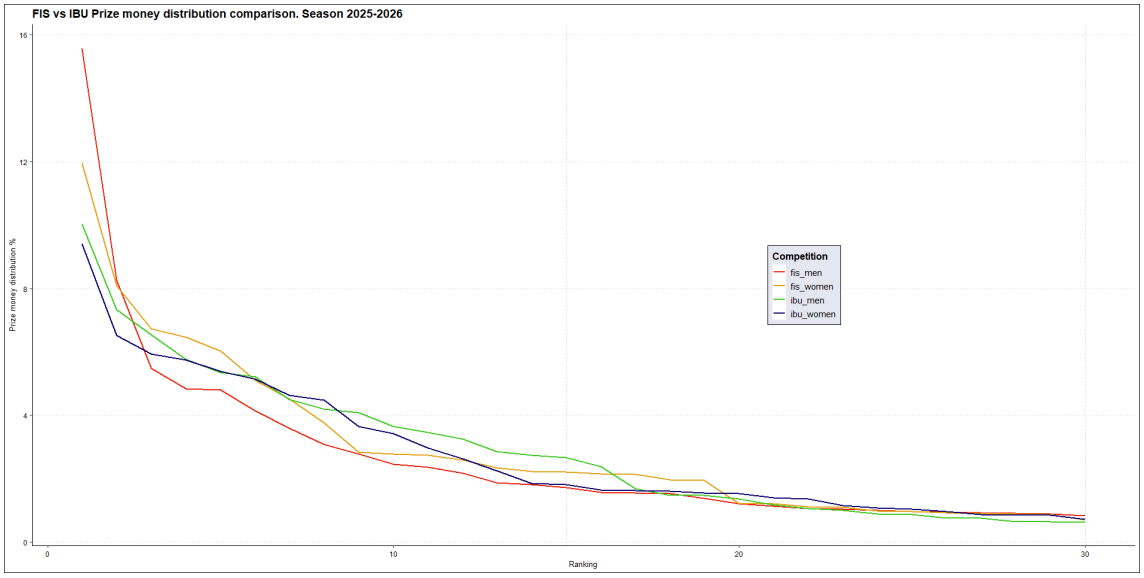

Figure 13: FIS World Cup vs IBU World Cup prize money distribution on top 30 most earned athletes in season 2025-2026

the sports with each other on this aspect.

As mentioned earlier in IBU World Cup 96 women and 95 men were granted some amount of prize money over the season 2025-2026. In FIS cross-country World Cup 88 women and 120 men were granted some amount of prize money over the season. Since the FIS World cup information is taken straight from the source, the information of total amount of athletes participating into competitions throughout the year is not available. It is however interesting to note that in men's competition there was way more athletes who were capable on finishing in top 20 in individual competition or top 10 in team sprints to get awarded by prize money.

In the emphasis of trying to make comparison slightly more visually understandable, the graph is cut into two pieces. First graph visualizes the top 30 athletes based on prize money earned and the second graph will visualize the rest.

Looking at top 30 athletes (13) there are clear differences visible. First of all the top athletes of FIS World Cup are taking a bigger cut from the whole prize pool. Especially in the men's side the line is very top heavy. There's a good reason for that as one athlete, Johannes Hoesflot Klaebo, is almost unbeatable and therefore he receives almost the maximum amount of prize money one athlete can win per season. This has to be taken into account, as it was already earlier mentioned that when top athletes takes bigger cut, there's less to share more. This does show especially in this case.

Women's distribution is managing comparison slightly better until around 7th or 8th athlete where the drop is more drastic. After this the drop ends and athletes from 9th to 19th seem to get fairly similar cuts from the prize pool and for brief period they perform the best. Top 20 to top 30 the differences starts to level out and all the lines start to follow the same kind of path.

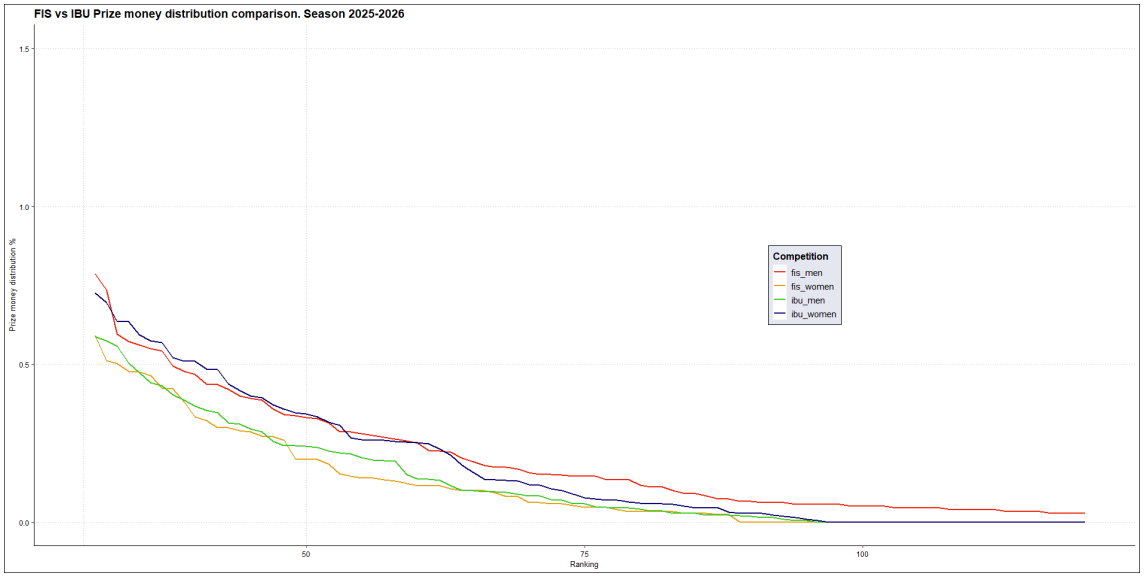

Figure 14: FIS World Cup vs IBU World Cup prize money distribution on the rest of the athletes who gained prize money in season 2025-2026

When looking the rest of the athletes from more closely there seems to be more differences but in grand scheme the differences aren't that big. It's good to mention that FIS World Cup's men athletes have bigger share for the whole time during this section compared to IBU's men but same can be said from IBU's women athletes in comparison to FIS's women. The major notice in this graph is that vastly wider range of men did manage to receive prize money in FIS World Cup than in other race formats. This is most likely due to higher competition divided into two different formats, sprint and distance athletes, and also two different skiing styles, classic and sprint, which was discussed about earlier. This is not as common in women's side which means that there are more women in the top who are highly competitive in sprint and distance competitions, with both styles.

When comparing the amounts of athletes who are capable of gathering a five figure prize money over the season, biathlon is performing way better compared to cross-country. in FIS World Cup only around 35 athletes in women and under 40 athletes in men were able to collect a five figure money. In this light it could be said that IBU was able to distribute it's bigger prize money pool for wider group of athletes in bigger chunks, but the difference especially in the men's competition isn't that big. In women's competition the difference is much more noticeable and this really shows also that IBU's distribution does work as long as competition is equal enough in the wider range of athletes. The reality with FIS cross-country World Cup is that the overall prize money awarded is not big enough that the clearly wider range of athletes in men's competitions capable on finishing in top 20 wasn't enough to widen the range of athletes who were able to collect significant amount of prize money throughout the season.

One big point is also to count how many percent of athletes who participated in the World Cup competitions throughout the year managed to receive prize money. As the FIS World Cup data set was collected straight from the public source, there isn't collected statistics from this ratio. In IBU World cup 55,8% of women athletes and 50,8 % of men athletes

manage grab their share of the prize money. This is fairly big part of all the athletes competing at least one during the season which can be seen as a positive result. The numbers for the FIS World Cup are not calculated due to only using the straight prize money share statistics from their official website.

8 Conclusion

Based on this analysis it's not yet completely possible to make definitive conclusions on either points distribution or prize money distribution effectivity. With further comparison it could be confirmed but changing the points distribution to award more points from top positions have made the overall competitions slightly less exciting at the end of the season as the differences between athletes are bigger. Of course it's good to mention that season 2025-2026 had Olympic Winter Games, which was the major goal for many top athletes but there was also some unfortunate events which effected the competition to overall victory in men's side. Having the smaller disciplines help with this challenge as these competitions are still tighter even at the end of the season. The latest changes on awarding athletes on the latter half of the top 10 with more points was a good addition.

There is a clear structure, a ladders to climb, for the athletes in the overall competition formats in IBU World Cup. There are 10 individual competitions where a whole pool of athletes are able to participate. There is a clear incentives for different levels. You have to be in top 60 best athletes in sprint to be able to compete in pursuit and add competitions in your calendar. It also gives you another chance to compete from points and even from more prize money. The next step is to get into top 40, in individual, sprint or pursuit, to earn points. Having top 30 to get awarded with prize money adds another step with something extra to compete from. in the end there's one more additional way to gather both points and prize money which is to perform in a high enough level to be able to participate in a mass start with top 30 athletes of given time but ideally athlete would like to be in top 25 in overall standings to quarantee this spot for every competition. Going closer to the top, athlete starts to compete from the individual podiums and wins, individual discipline wins and eventually overall Total Score win. The overall structure is very good and doesn't require major modifications. The relays give opportunities for the most of the athletes in the circuit to participate more competitions throughout the season.

In general IBU is also doing fairly good job with prize money distribution and proposals for the current system with the same pool of money are very limited. The amount of prize money given out has been rising year by year and as this is expected to be continuing it will be interesting to see how it will added to the current system. Based on this analysis some reasonable proposals could be made.

- Keeping the current prize money distribution for future year with IBU World Championships would be an absolute treat but likely not possible.
- If going back closer to the earlier years distribution, the emphasis could be trying to improve the tail end closer towards the this years distribution. Only after the tail is improved to this seasons standard, the winner's prize should be improved.

- Pursuit and relay leg bonuses were a great addition. The prize range is also good.
- Relays could award top 10 teams as in FIS team sprints.
- Total Score bonuses could reach up to top 15 athletes. Keeping the five figure bonus mininum is still more meaningful compared to the FIS World Cup comparison.
- Small disciplines could have top 3 athletes granted with a prize money instead of only winner, including the U23. This would mostly grant the top end but would give also some suprise athletes a possibility to gain a small extra at the end of the season.

Most of these proposals expect only marginal improvement on prize money pool. Keeping the season 2025-2026 level would require about 1,5 million euros added to the pool which most likely is not possible. In general the top athletes were able to collect a very hefty amount of money and therefore it would be good to aim to contribute most of the new money to the mid tiers at least for now. It's would be also nice to see if all the athletes who manage to end up on points, would be granted some prize money as well but as seen on this analysis, this wouldn't add up too many athletes to group of benefitters. It would most likely however improve the prize money gathered of the bottom half of those who already were granted some during this year which would improve the healthiness of the scene. Even with quick calculations this would also require quite a large sum of additional money.

In FIS cross-country World Cup the distribution is a lot more top heavy even disregarding the Kläbö dominance in men's overall competition. The improvements were done for it already in this year but it still doesn't feel good that third place finisher is getting less than 50% of the winner's prize money (read comparison in chapter 6). This part of the distribution got the major improvement compared to earlier years. In a way FIS is working with a completely different challenges in this regard as it would need to still focus on adding major improvement for 2-6 places in terms of prize money to make it feel much more worth while before even discussing of adding more placements who will earn prize money altogether. Overall bonuses at the end of the season for top 20 is relatively nice but the amounts after top 10 are so small that they are almost insignificant buff to what athletes on these positions have been already earning during the year. Even distributing money to wider group is better, it still feels like IBU is making better job with this part of the distribution as well. Distribution in FIS World Cup still needs a lot of improvement to be more healthy even the prize money pool wouldn't increase to the levels of IBU related events.

Viitteet

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